Statistical Learning Assignment 2

Petridou Elena (s2029197)

May 2025

Part A

Question 1: Select one supervised learning method from those that were discussed in weeks 7 and 8 for analyzing this dataset. The chosen method must be different from the ones in Assignment 1! Justify why you would select this method for this specific prediction problem. (Use max. 200-250 words.)

An important consideration in model selection is its interpretability, especially as we are asked to assess feature importance in this assignment. Support Vector Machines (SVM), are interpretable with linear kernels, but become difficult to interpret as the kernel function becomes more complex. Given that our dataset includes categorical and numeric variables, it is likely that a non-linear kernel will be most suitable, thus limiting interpretability. Despite this, there are post-hoc methods (such as SHAP-values or permutation importance) available for gleaning feature importance in non-linear SVMs, so this is not the only consideration.

Further, there also appears to be interaction between the covariates and the outcome, as seen in Figure 1. Tree-based methods are more flexible in modelling interactions. Friedman explains that "A tree with J terminal nodes produces" an estimate function "with interaction order at most min(J-1,n)" (2001, p. 1215). Thus, the most suitable tree size depends on the order of the interactions in the data-producing function in the real world. "This is usually unknown so that J becomes a metaparameter of the procedure to be estimated using a model selection criterion such as cross-validation." (2001, p. 1215). In the XGBoost r library, J is determined by *max_depth* which can be tuned using cross-validation. On the other hand, SVMs capture interactions or complex relationships in the data via the kernel shape (linear, polynomial, radial). Kernel shape, however, is not one of the hyperparameters tunable via cross-validation, meaning that we would have to select it from beforehand, or compare models with different kernels. Since the order of the interactions is hard to evaluate a-priori in order to pick a suitable kernel for SVM, ensemble trees via XG-boost offer a more elegant solution to accurately capturing the interactions in the data.

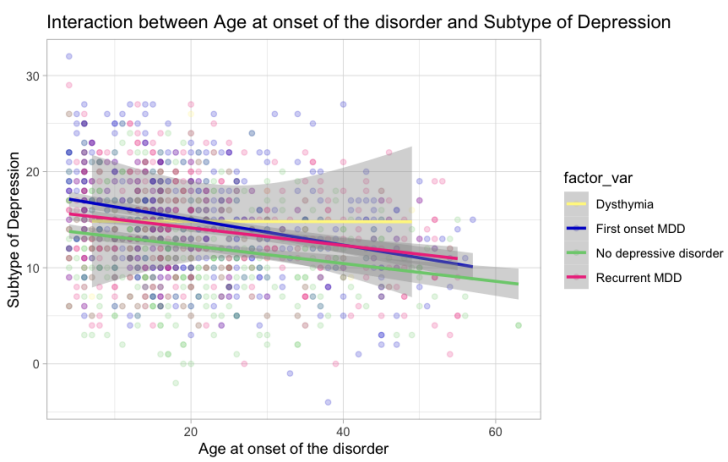

Figure 1: Interaction of IDS score and disorder type (on outcome)

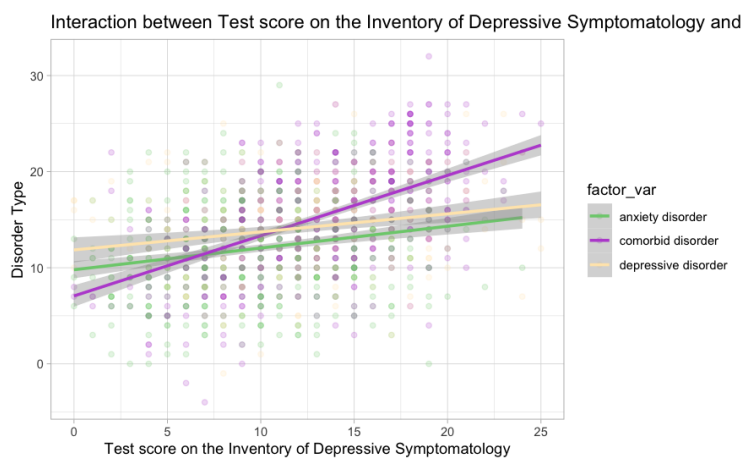

Figure 2: Interaction of age at onset and disorder type (on outcome)

Question 2: Now apply this method you selected to the dataset. Beforehand, randomly split the dataset into a training (n=1000) and test (n=500) dataset. Use your student number to set the seed of the random number generator.

Question 3: Justify your choice of the main model-fitting parameters. Thus, make a well-informed choice for a fixed value of each parameter and/or use cross-validation to set their values. Your answer should reflect understanding of what each parameter does. (Use max. 200-250 words.)

To train gradient-boosted trees, I use the computationally-efficient XGBoost algorithm. I opt for hyperparameter-tuning using cross-validation via the Caret library, and using a search-grid made up of values which I justify in this section. The aim is to ensure that the model does not overfit the data (introduce bias to reduce variance), while still

fitting enough trees to capture patterns in the data sufficiently (keep bias reasonably low). This leads to the learning rate (ν) versus number of boosting rounds (M) trade-off (referred to as $\nu - M$ trade-off in Friedman, 2001 (p. 1206).

M is given by *nrounds*, which Friedman suggests should be made as large as computationally feasible, but performance improvement for which appears to "flatten out" at approximately $M \geq 200$ (p. 1206). As such, I choose 200.

Step-size shrinkage (*eta*) controls the learning rate. Higher values introduce less bias to the model, allowing for more "dramatic" adjustments of the weights of the model between every sequential tree-fitting. As noted by Friedman, smaller values of this shrinkage parameter, "provide higher accuracy, with a diminishing return for $\nu < 0.125$" (2001, p. 1206). Thus, I chose the range of values: (0.01, 0.05, 0.1, 0.2, 0.4).

Thereafter, other hyperparameters can also be tuned.

The maximum depth of a tree (*max_depth*) can be adjusted to make trees shallower (which avoids over-fitting), or deeper (which allows for more faithful capturing of the patterns in the training data – including the interactions). I specify a grid of conservative values for tree depth, ranging from 3 to 7.

Gamma is another hyperparameter which can make the model more conservative, thus reducing overfitting. It prevents the tree from splitting unless a threshold of loss-reduction would be achieved by the next split. I specified values (0, 5, 10, 20).

To pick the most suitable hyperparameters I then use 10-fold cross-validation and extract the best hyperparameters, which are shown below, along with their performance metrics:

Table 1: Best Tune Hyperparameters

nrounds	**eta**	**max_depth**	**gamma**
200	0.05	3	5

Table 2: Error Estimates for Best Tune Model

Error Estimate	**Value**	**Standard Deviation**
RMSE	4.632373	0.2702807
R-Squared	0.2391655	0.06427721
MAE	3.728518	0.2007246

Question 4: Provide an interpretation of the resulting model:

Describe which variables are most important in predicting the value of the outcome variable, and which measure(s) you used to determine their relative importance.

The ten most important features to the prediction are shown in the table below, along with three metrics for feature importance. The sum of each metric, across all features is 1, meaning that we can compare the features to each other.

Gain quantifies how much the model improves when this feature is used to inform a split. It represents the error reduction (MSE) when this variable is used to split, averaged across all trees. The feature with the highest Gain is a patient's test-score on the Depressive Symptomatology Inventory (IDS). It accounts for almost a quarter of the total gain that features contribute to the model.

Cover quantifies what proportion of subjects/entries this variable affects in its splits across all trees.

Cover and frequency are very similar, except frequency counts the proportion of splits that this feature is involved in. For example, IDS has a high cover value (0.114) – meaning that when the tree splits on IDS, many subjects are "affected". This likely means that the splits IDS is involved in might be higher in the tree, or create large groups. Whereas this same variable has a low frequency score (0.098). This means that IDS is not used as a splitting criterion in many trees, but when it *is* used, then it is quite informative. Since IDS has high gain and high cover, this is clearly the most important feature in predicting depressive symptoms after twelve months.

The other four features making up the top five features are: Age at disorder's onset, percentage of time that symptoms have been present in the past four years, the patient's age, and the type of disorder (anxiety, depression, or both) that a patient suffers from. All five of the most important features have relatively high gain and relatively high cover. Though features with a higher number of values (e.g.: numeric instead of categorical variables) can tend to have higher frequency or cover, given that all the top variables have *both* high gain and high cover shows that they are informative and the model does not only prioritise them for splits simply because of the scale of their values.

Table 3: Error Estimates for Best Tune Model

Feature	Gain	Cover	Frequency
Testscore on the Inventory of Depressive Symptomatology	0.258641818	0.114376494	0.09808696
Age at onset of the disorder	0.122299024	0.145823122	0.12243478
Percentage of time with symptoms in the past 4 years	0.106405024	0.150440895	0.14052174
Age in Years	0.103630130	0.108535817	0.14747826
Type of disorder (depressive disorder, anxiety disorder or comborbid disorder)	0.076400684	0.018396071	0.04556522
Testscore on Beck's Anxiety Inventory	0.074601068	0.096480589	0.09947826
Testscore on the Fear Questionnaire	0.061195191	0.101305931	0.09286957
Years of Education Completed	0.046095379	0.044023278	0.06469565
Diagnosis of Generalised Anxiety Disorder	0.023154165	0.021878470	0.01808696
Subtype of Depression	0.021340352	0.022318615	0.02539130

Describe the effect of the most important variables (e.g., describe the shape and direction of the effect on the predictions and/or provide and discuss plots of the variables' effects). (Use max. 150-200 words)

Figures 2 and 3 show a summary plot of the effect of the top five most important features on the outcome prediction. These are computed using SHAP, a post-hoc feature evaluation method based on Shapley values. SHAP assesses the impact of a feature on the predicted outcome for each entry, then SHAP values can be aggregated across all entries in order to assess the total influence of a variable on the whole model.

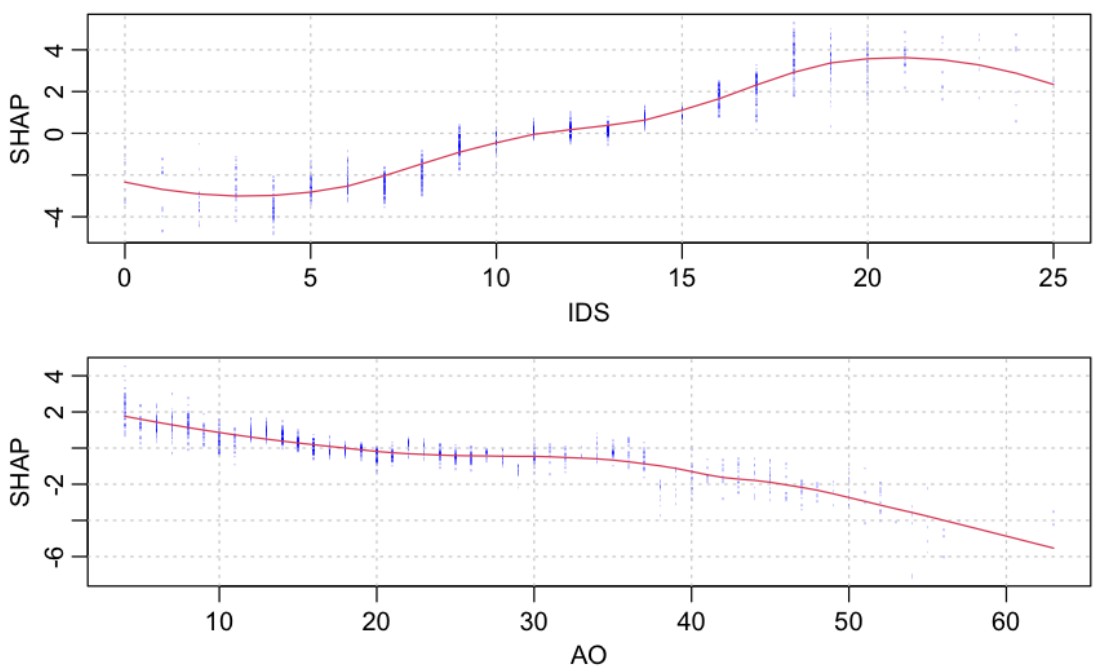

Figure 3: SHAP values for the top two features

As we can see from these plots, higher scores on the Depression Symptomatology Inventory lead to higher predicted values of depression severity at the end of the study – while holding all other variables constant. An IDS score higher than 15 indicates that the predicted severity after 12 month will be higher than baseline, whereas a score of lower than 10 indicates that the prediction will be lower than baseline.

The opposite is true for Age at Onset: the younger a patient is at the age of onset of their symptoms, the higher the predicted value of their outcome. If a patient is younger than 20 then their predicted outcome is higher than baseline, whereas if they are older than 30 their predicted outcome is lower.

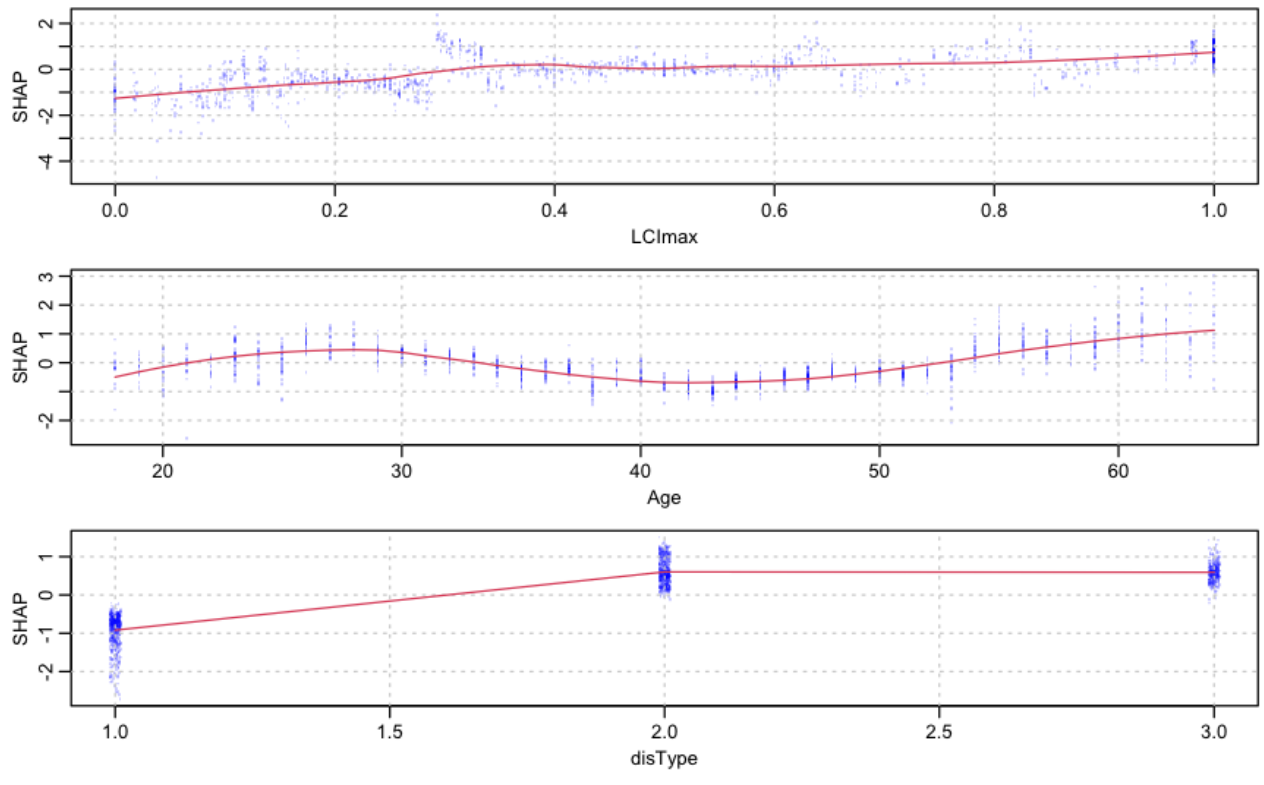

Figure 4: SHAP values for the next 3 most important features

For current age of the patient, the trend is not so clear. An age in the mid-twenties seems to lead to a higher than baseline predicted value for the outcome, whereas age between 40 and 50 leads to lower than baseline. For ages greater than 55, however, the predicted value of the outcome is higher than average.

The relationship between the percentage of time of symptom presence and predicted depression severity at 12 months is not as drastic as for the other features. However, if symptoms were present for less than 40% of the past 4 years, the predicted value is lower than average.

Assess the predictive accuracy of the model using the test set.

The final model performance is very similar to the best-tuned training model. The results of the same tests are tabulated below.

Table 4: Error Estimates for Final Model (on the Test set)

Error Estimate	**Value**
RMSE	4.413379
R-Squared	0.241506
MAE	3.548307

Question 5: Now create a Table comparing the two models in your Assignment 1 and the new model fitted in Assignment 2. How do the predictive accuracies compare? Which model predicts best? (Use max. 200 words)

Table 5: Error Estimates for Final Model (on the Test set) for the three models

	XGBoost Tree Ensemble	**Conditional Inference Tree**	**Lasso Regression**
RMSE	4.413379	4.5269471	4.2931063
R-Squared	0.241506	0.1970624	0.2761241
MAE	3.548307	3.6227565	3.4424642

The Lasso Regression outperforms all other models, with both a lower Root Mean Squared Error (RMSE) value than the other models, and a higher R-squared value. The difference in both of these metrics between LASSO and XGBoost is small, with an R-squared difference of approximately 0.0346 and RMSE difference of about 0.1203, showing that either of these models could be a reasonable choice.

The fact that LASSO outperforms the other models shows that in a dataset such as this one, with many (correlated) variables, the feature-selection carried out by Lasso is valuable, as it produces a model with only the most relevant predictors.

None of the models capture a very large proportion of the variation in the outcome, with the best R-squared value produced by LASSO being approximately 0.276 – meaning the model is able to explain less than 30% of the variability. However, this is a rather large effect size in the field of Psychology (Funder & Ozer, 2019), meaning that LASSO is a rather good model for this particular problem. The RMSE is also reasonably low, at approximately 4.293.

Question 6: Based on 3-5: Provide a short overall conclusion regarding which predictors are related to the outcome and which model you would recommend and why.

Below, I tabulate the most important predictors from the best-performing model (LASSO) – which I re-ran in order to ensure reproducibility. As we can see, some of the most important features in this model overlap with the most important features identified by the Gradient-boosted ensemble trees.

Table 6: Coefficients - LASSO

Variable	Coefficient
Type of Disorder: Anxiety (disType)	-1.49812921
Percentage of Time symptoms were present in past 4 years (LCImax)	1.38015266
Antidepressant Use (ADuseTRUE)	-0.98801816
Positive Generalised Anxiety diagnosis (bGAD)	0.97573451
Patient Receiving Psychiatric Treatment (PsychTreatTRUE)	-0.72267876

Percentage of time symptoms were present in the past four years is the most important predictor in LASSO regression, and is also within the top five most important features for the ensemble. Since both models corroborate that this is an important feature, we know that its importance is less likely to be due to technicalities of the model. We are interested in predicting depression symptomatology progression at a follow-up of 12 months, thus it is reasonable to assume that presence of symptoms for a significant proportion of time before would be a useful predictor when assessing whether symptoms will persist in the future.

Disorder type is also an feature in all three models – the Conditional inference tree used it as its first split, it is in the top-5 gain features for XGBoost, and is the fourth most important feature in LASSO. Indeed, a patient's diagnosis is reasonably expected to predict how severe depression symptoms at follow-up will be. As we can see from the coefficients table, having anxiety disorder leads to lower symptoms estimate at follow-up than patients who suffer from depression or both.

The other top features were not shared between the models. For LASSO, antidepressant use and receiving psychiatric treatment are important predictors for symptom severity at follow-up. However, these variables have negative coefficients, meaning that patients taking anti-depressant medications or receiving psychiatric treatment on average have lower predicted depression symptoms at follow-up.

Seeing as the best performing model (LASS0)'s performance is not vastly better than the gradient-boosted ensemble trees' the top features for XGBoost can also reasonably be assumed to have good predictive value on depression severity at follow-up. Thus, Testscore on the Inventory of Depressive Symptomatology and Age at onset of the disorder might also be good features to consider if investigating this question in future studies.

As explained in the previous sub-section, the most suitable model for this problem seems to be LASSO regression, and this is likely due to the fact that this dataset contains numerous predictors. Thus, the feature-selection performed by LASSO is essential in order to use only the most relevant features in the model.

Question 7: A psychologist has seen David Edgar Pression for an intake today. The psychologist wonders whether they should refer David to an intensive depression treatment program. The results of David's intake assessment are provided on Brightspace, in the file 'Patient.csv'. The psychologist asks you to provide them with an estimate of the severity of David's depressive symptoms in 12 months. Patients with predicted depressive symptom severity equal to or greater than 17 are referred to the intensive treatment program. What is your estimate? Should David be referred to the intensive treatment program? Bonus: Using a suitable approach, quantify the uncertainty of your estimate (not specifically taught during the lectures). (Use max. 100 words, max. 200 words including bonus.)

The best predictive model (LASSO) yields an estimate of 16.53648 for David. As such, David should probably not be admitted to the intensive programme as patients with a score of 17 or higher should be admitted.

Part B

Question 1: Do you think that reducing the variables to a smaller set of (new/derived) variables may be a good idea? Which statistical technique can be used to achieve such a dimension reduction? Explain why this technique is appropriate.

The dataset contains thirty variables (features) investigated for 1000 individuals (entries). This leads to a relatively sparse dimensional space, as the ratio of entries to features is rather low. Since there are so many predictors, dimensionality reduction could reduce problems such as the curse of dimensionality which some models could suffer from.

Moreover, reducing the number of features is a good idea when features are correlated with each other, as the variation explained by each feature can be approximated by its highly-correlated counterparts. Reading through the personality traits in the dataset, we can realistically expect that some of them are likely to be correlated with each other. For example, gregariousness, competence, and achievement striving can be expected to co-occur in people. On the contrary, we might expect that depression can be negatively correlated with positive emotions, or angry hostility to be negatively correlated with trust. We can check these assumptions with a correlation plot – reported in Figure 5.

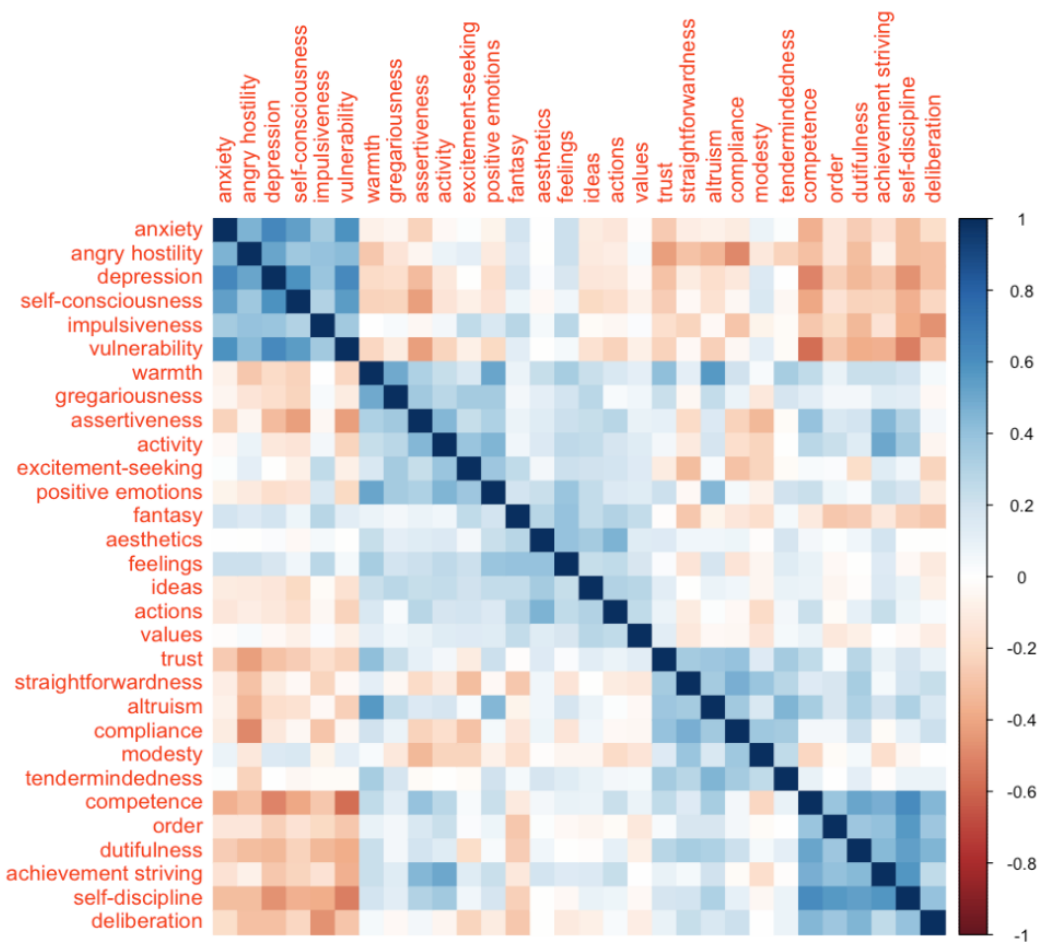

Figure 5: Correlation Plot for all Characteristics

As such, dimensionality reduction could combine some of these features into new, uncorrelated representations which still represent the original data well. A suitable approach for this is Principal Component Analysis (PCA): an unsupervised dimensionality-reduction technique which linearly combines features into principal components. Its aim is to transform the features into orthogonal principal components that maximally capture the signal in the data. Orthogonality of the principal components essentially guarantees that they are uncorrelated. This is a suitable procedure as information density is relatively well-preserved, and the feature space is reduced such that interpretation of the model becomes easier.

Question 2: How many new/derived variables should be computed to capture the most important part of the information in the original variables? Use at least four different methods to select the number of new/derived variables. Explain each method and thoroughly justify your final decision (e.g., by providing relevant figures).

The first method which can help determine how many principal components should be retained is inspecting Scree plots which show how much variance in the dataset each individual component captures, as well as how much of the cumulative variance in the whole dataset is captured by the principal components computed thus far. Since the more

principal components we use, the more of the total variance we will explain regardless of whether or not the components are good approximators, we should try to find the smallest number of PCs which explain a high-enough cumulative proportion, and ensure that each component explains "enough" variance so as to not be redundant. This approach was first suggested by Cattell (1966), and involves inspecting the graphs for the sharpest drop in the slope of decreasing variance explained, before the line tapers off – a so-called "elbow" in the graph. As we can see in Figure 6, the elbow for the variance explained by each component lies at about 4, but 6 is also a fair answer. As for the cumulative variance explained, the graph is smoother so no clear "elbow" can be observed. However, the graph seems to begin tapering off at 3 or 4 components, where it captures about 50% of the total variance in the dataset. By 6, the PCs capture 60% of the variance. Ledesma et al. (2015) criticise this lack of clarity as to the actual last pronounced slope in graphs, illustrating that the Scree plots in such situations are not the most suitable method for determining the correct number of principal components. For this reason, it is important to review multiple methods and pick the most appropriate number by combining the information from all of them.

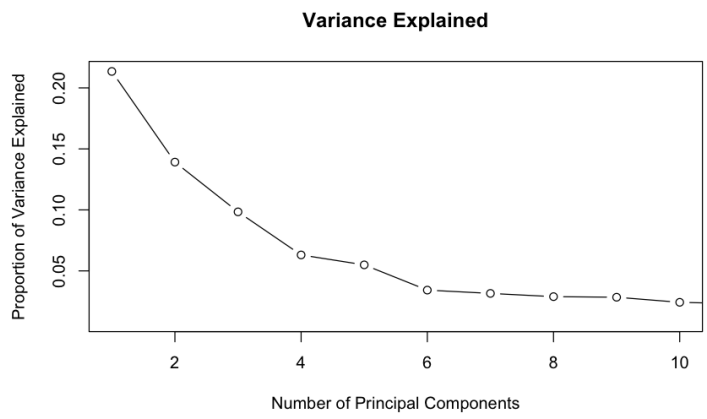

Figure 6: Scree Plot for Variance Explained by each Principal Component

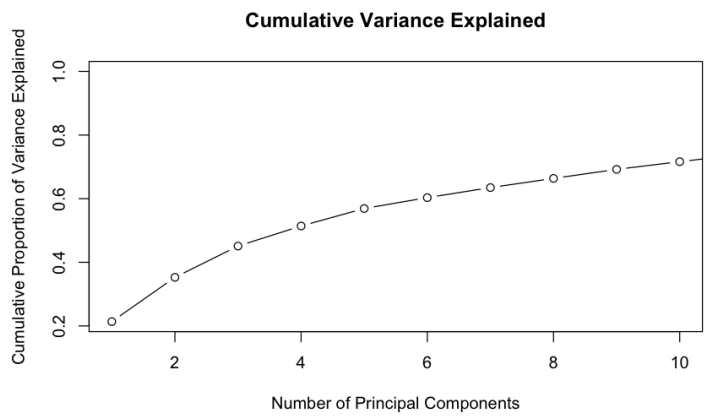

Figure 7: Scree Plot for Cumulative Variance Explained by this Number of Principal Components

A second method which can be used for whittling down the number of principal components is Horn's Parallel Analysis. This method is related to Kaiser's rule (Kaiser, 1959 as cited in Horn, 1965, p.181), who states that principal components with an "eigenvalue greater than one" carry significant information about the dataset to make them valuable for retaining. This is the rule which also underpins the logic of the Scree plots we saw above. However, Horn argues that the correlation in the data is not attributable only to the data itself (true-correlation) – as assumed by the previous method – but can also arise due to "sampling error and least-squares bias" (1965, p. 181). His method uses simulation to estimate what the average eigenvalues would be for completely random matrices possessing the same dimensions as our dataset. Using these estimates, we can "deflate" the eigenvalues of our empirical principal components, producing more accurate estimates. The principal components with adjusted eigenvalue greater than 1 can be considered good approximators for the original dataset. In Figure 8 we see that the number of principal components which we should consider keeping using Horn's parallel analysis is five.

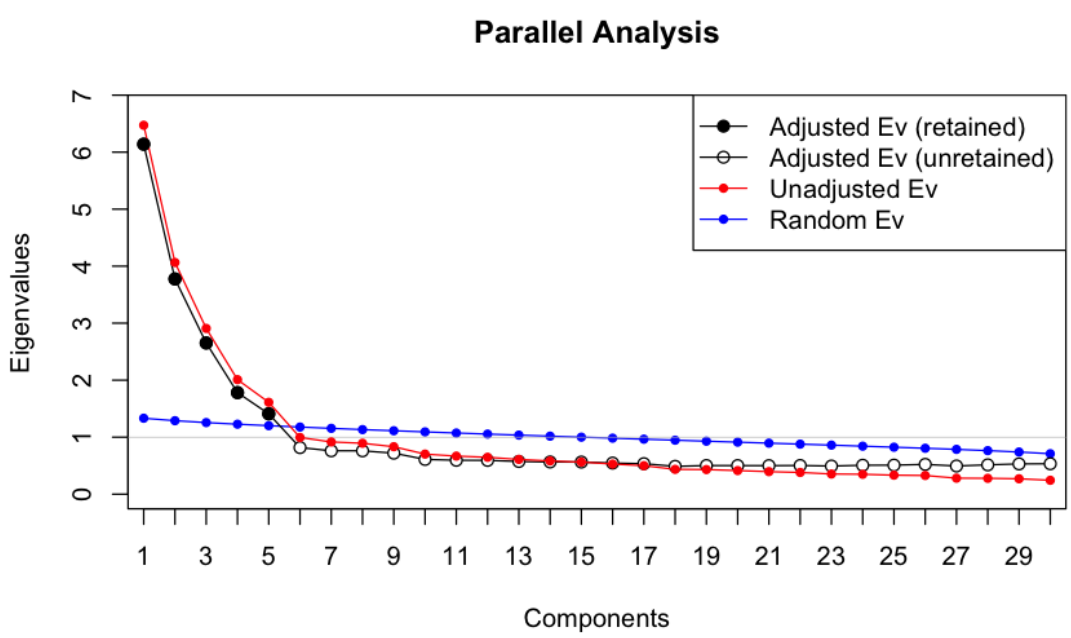

Figure 8: Horn's parallel analysis graph

```
Very Simple Structure
Call: vss(x = cor(US_data), n = 15, rotate = "none", diagonal = FALSE,
    fm = "pc", line = TRUE)
VSS complexity 1 achieves a maximimum of 0.59  with  3  factors
VSS complexity 2 achieves a maximimum of 0.77  with  3  factors

The Velicer MAP achieves a minimum of 0.01  with  5  factors
BIC achieves a minimum of  Inf  with    factors
Sample Size adjusted BIC achieves a minimum of  Inf  with    factors

Statistics by number of factors
```

Figure 9: VSS and Velicer's MAP

A third method is the Very Simple Structure (VSS) criterion, proposed by Revelle and Rocklin (1979). What the VSS does is compare the full-component matrix sequentially with simplified versions. The process begins by extracting the first say k components and simplifying them such that each variable only loads high on one of the components (thus, if, for example, anxiety loads high on PC1, then we zero out the other loadings such that anxiety *only* loads on PC1 and is the only non-zero loading on PC1) – as such, VSS sets anxiety to 0 for the other principal components and the other features to 0 for PC1. Then, this simplified correlation matrix is compared to the full correlation matrix by calculating the VSS score: 1 - Sum of Squares (Simplified Matrix)/Sum of Squares (Full Matrix) (RDocumentation, n.d.). The test continues by extracting k+1 components and repeating the aforementioned procedure. The maximum value of the VSS score then shows how many principal components should be extracted, as this amount maximises the explained variance. The fact that the VSS favours principal components with certain features loading high only on some components means that it favours more interpretable solutions, while also trying to maximise how much of the original variance in the data the principal components capture. The output of the VSS function is shown below, and indicates that according to the VSS criteria, the most suitable number of components to extract is 3.

The fourth method, Velicer's minimum average partial (MAP) test is also reported in the above table. This method functions in a similar way as the VSS, in that it compares the full correlation matrix to reduced correlation matrices, but this time it does so by removing certain components from the matrix. Then, it compares the average partial correlation between variables after removing a number of components. When this number reaches a minimum, it indicates the ideal number of principal components which we should extract as this number of components factors out the most feature correlation from the data. This can lead to less interpretable results than VSS, as there is not as much emphasis on ensuring that the remaining components have distinct loadings. This is especially evident in our case, where MAP recommends 5 principal components as opposed to the 3 recommended by VSS.

Question 3: Can you give a meaning to these new/derived variables?

The tables below show the top loadings (most positive and most negative) of each extracted principal components.

Principal Component 1

The first principal component has high loadings for traits which seem to correspond with high confidence and high achievement, such as achievement striving and competence. It has low loadings for traits associated with emotional turbulence, such as impulsiveness and angry hostility. This seems to indicate whether a participant has a stable temperament and prone to positive interactions with others, shown by altruism and warmth.

	PC1 <dbl>
competence	0.2936347
self-discipline	0.2893141
dutifulness	0.2437916
achievement striving	0.2300190
altruism	0.2057086
warmth	0.2004855

Figure 10: Most Positive Loadings

	PC1 <dbl>
vulnerability	-0.2896978
depression	-0.2794867
self-consciousness	-0.2477055
angry hostility	-0.2323627
anxiety	-0.2235034
impulsiveness	-0.1746862

Figure 11: Most Negative Loadings

Principal Component 2

The second principal component indicates whether an individual has a high degree of straightforwardness, modesty, and modesty, and negative loadings on feelings, excitement-seeking, fantasy, etc. We might characterise this component as a person being "down-to-earth" and serious; unlikely to engage in new activities or spontaneity. What is interesting about this component is that the most postive loadings are not as balanced in their scores as they were for the first principal component, whereas the negative loadings are quite well-balanced. This shows that this component refers more to someone who exhibits low levels of "excitement"-related characteristics.

	PC2 <dbl>
straightforwardness	0.24475509
compliance	0.21959773
modesty	0.20119900
deliberation	0.20059237
dutifulness	0.12777437
order	0.07402574

Figure 12: Most Positive Loadings

	PC2 <dbl>
feelings	-0.3083799
excitement-seeking	-0.3077496
fantasy	-0.2852850
activity	-0.2643629
positive emotions	-0.2626037
assertiveness	-0.2345129

Figure 13: Most Negative Loadings

Principal Component 3

The third principal component seems to refer to a personality trait encompassing softness, tendermindedness, compliance and altruism. Assertiveness, achievement striving and self-discipline are this component's most negative loadings. This may indicate the extent to which someone is warm and giving nature averse to hostility or strict regiments owing to the negative loadings of order and competence. In contrast to principal component 2, this principal component has rather high positive loadings and rather low but equally balanced negative loadings.

	PC3 <dbl>
tendermindedness	0.3749569
compliance	0.3349432
altruism	0.3063174
modesty	0.3004105
warmth	0.2824978
trust	0.2644782

Figure 14: Most Positive Loadings

	PC3 <dbl>
assertiveness	-0.1630408
achievement striving	-0.1323780
self-discipline	-0.1233047
angry hostility	-0.1192682
order	-0.1155039
competence	-0.1100733

Figure 15: Most Negative Loadings

Principal Component 4

This principal component seems to idealism as values, ideas and fantasy are its three most positive loadings. This loading indicates also low order, anxiety and achievement striving. The most positive loadings are not balanced in their distribution of scores, as values, ideas, fantasy, actions and trust have much higher scores than the last – compliance, which has half the weighting score of the next top-loader, trust. The negative loadings all have quite a high absolute value and relatively balanced scores.

	PC4 <dbl>
values	0.23704519
ideas	0.21395363
fantasy	0.18042938
actions	0.17959648
trust	0.16251470
compliance	0.08766565

Figure 16: Most Positive Loadings

	PC4 <dbl>
order	-0.3575805
anxiety	-0.3260257
achievement striving	-0.2968096
self-consciousness	-0.2603639
angry hostility	-0.2525449
activity	-0.2398438

Figure 17: Most Negative Loadings

Principal Component 5

The last extracted principal component shows attunement to aesthetics, action, deliberation and fantasy, with its most negative loadings being on gregariousness, excitement-seeking and positive emotions. This could indicate the extent to which a person is a "mysterious artist" type – more prone to creative pursuits than extraversion.

	PC5 <dbl>
aesthetics	0.4564305
actions	0.4355600
deliberation	0.2237323
fantasy	0.1916969
feelings	0.1808279
dutifulness	0.1759231

Figure 18: Most Positive Loadings

	PC5 <dbl>
gregariousness	-0.3167743
excitement-seeking	-0.2427579
positive emotions	-0.2154213
altruism	-0.1819227
warmth	-0.1710404
impulsiveness	-0.1606138

Figure 19: Most Negative Loadings

Question 4: Can you somehow quantify the degree to which the new/derived variables capture the information in the original variables?

Yes, since the loadings for each principal component indicate how much a feature contributes to that principal component, squaring the loading contributions (to get rid of the negative values), and then plotting the sum of these squared loadings in a barplot, shows us to what extent each feature is represented in *all* the principal components. What we see here is that not all features contribute equally. Activity, for example makes up more than 0.25 of the variation in the total squared loadings in the five principal components, as opposed to, for example, competence, whose total squared loading magnitude is about 0.12.

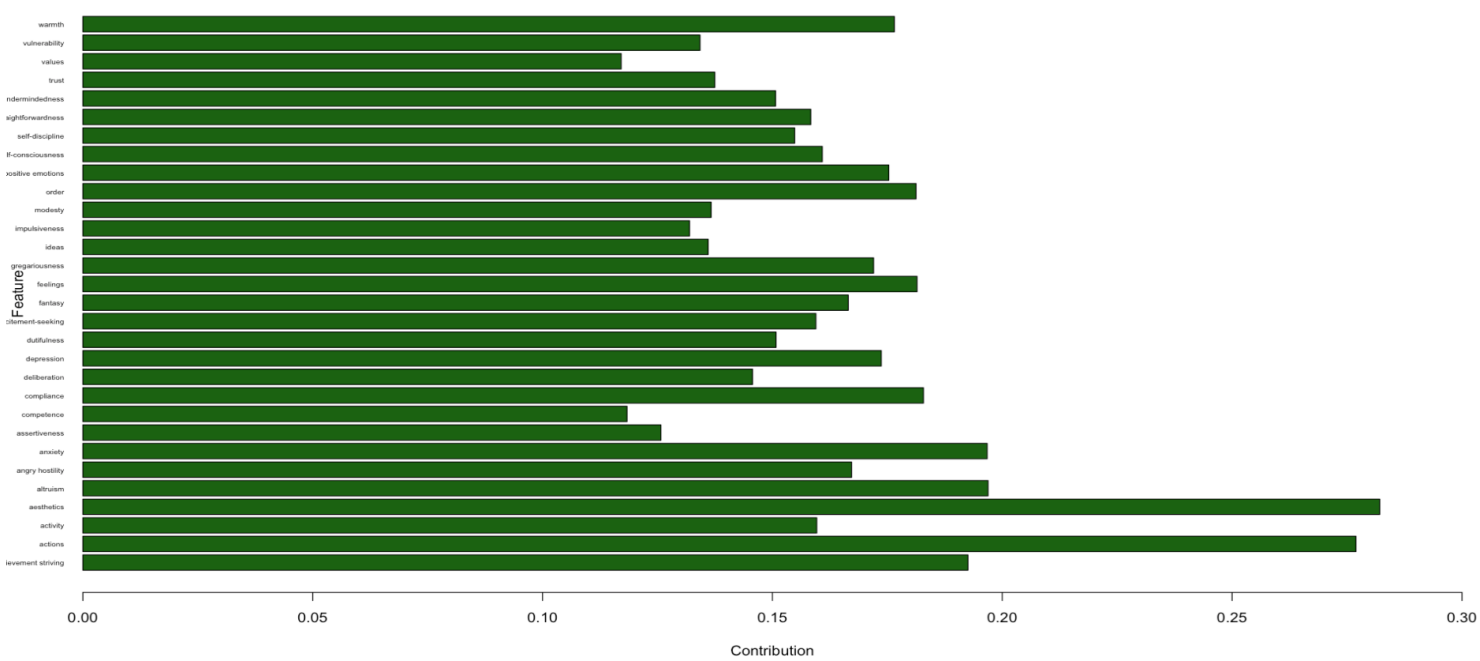

Figure 20: Feature Contributions to the 5 principal components

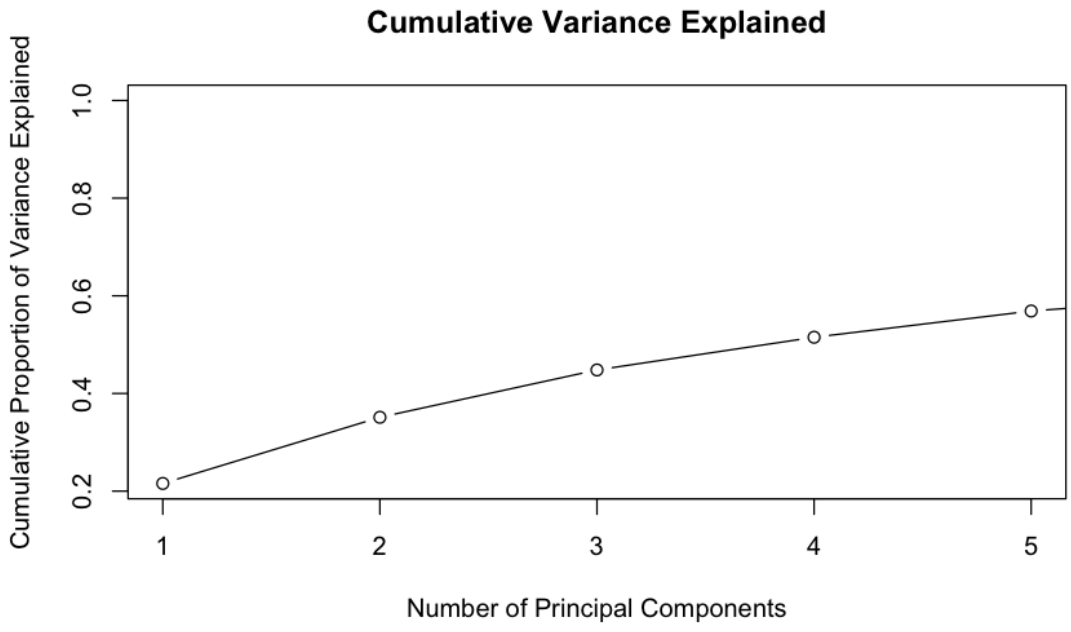

Figure 21: Cumulative Variance in original data captured by 5 Principal Components

Another way to visualise how much of the information in the original variables our principal components capture is via the cumulative variance graph shown below. It indicates that five principal components capture approximately 60% of the total variance in the original dataset.

Question 5: Which statistical technique(s) can be used to group the participants? Explain why it is/they are appropriate.

Since this is an unsupervised learning problem, seeing as we have no outcome variable, and since we are trying to group participants into within-similar and between-different groups, we can use clustering methods to achieve this aim. Clustering techniques seek to group datapoints into clusters such that the datapoints within the same cluster are as close as possible to each other.

The simplest method of doing this is k-means clustering, whereby we specify a certain number of desired clusters, and the algorithm tries to assign clusters to the data such that the within-cluster variation is minimised for all groups. This technique is not so well-suited to our problem, as we are not sure how many groups of personality-types we wish to extract, and we are also not sure what the size of each group would be. Since this is a centroid-based method, it does best on applications where groups are assumed to have similar sizes.

Another way of performing clustering is hierarchical clustering: this method attempts to assign the data to a suitable number of groups based on which number of partitions best maximises the within-group homogeneity and between-group heterogeneity. Hierarchical clustering is usually used when we are not sure what number of groups we wish to split the data into. This is a more suitable approach for our problem, since the dendograms produced by such a method can help us determine which is the most appropriate number of personality groups in this dataset.

A third way of performing clustering is gaussian-mixture clustering, whereby every group is assumed to have come from its own gaussian distribution with a different mean and standard deviation. Then, each datapoint is assigned to one of these groups based on how probable it is that this datapoint arose from each group's distribution. This approach handles groups of different sizes better than the k-means clustering method, meaning that it is more appropriate for our set-up. It also allows us to choose the most suitable number of groups based on a test statistic.

Question 6: How many groups are there? How did you determine this? Thoroughly justify your answer (e.g., figures). If you identified more than one technique in the previous question, choose two and compare the techniques and the obtained results.

All three techniques from above were used. Unfortunately, gaussian-mixture clustering ended up recommending just one cluster. Initially, I assumed that this could be due to the dimensionality of the data making it difficult to tell apart clusters, with 5 principal components. However, attempting to use fewer did not alleviate the issue. This could indicate that there are no clear clusters in the data based on any of the principal components specified, as they do not arise from different gaussian distributions, or it could mean that this method is not well-suited to clustering in this particular dataset.

I proceed with selecting groups based on the other two procedures. First, running k-means clustering for 10 possible groups and plotting the within-cluster sum of squares achieved in each k, I observed an "elbow" appearing at k = 2. However, the within-cluster sum of squares reduction appears to reduce definitively linearly only after k = 5. This

could indicate that either 2 or 5 groups are a good idea. Since this is not a very concrete way of determining, I looked to the results obtained via hierarchical clustering next.

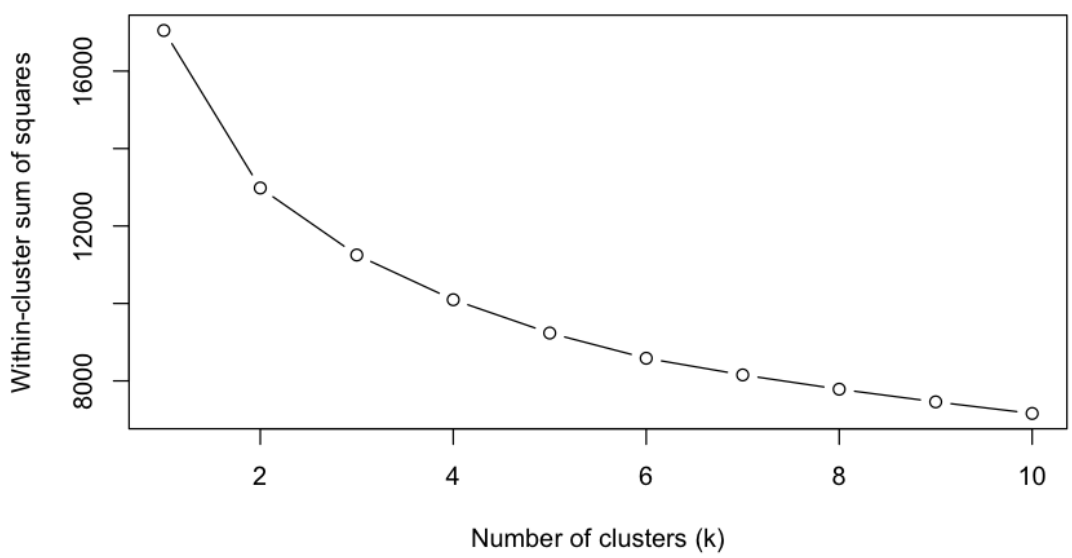

Figure 22: Within-cluster sum of Squares for k: 1-10

First, attempting hierarchical clustering with complete linkage results in the plot in Figure 23. Complete linkage is a dissimilarity measure which aims to maximise the single dissimilarity between groups (Lovik, 2025). The dendogram plot below shows that the most suitable number of most-dissimilar groups is 78. This is too high a number of groups to be interpretable. As such, I try a different linkage method.

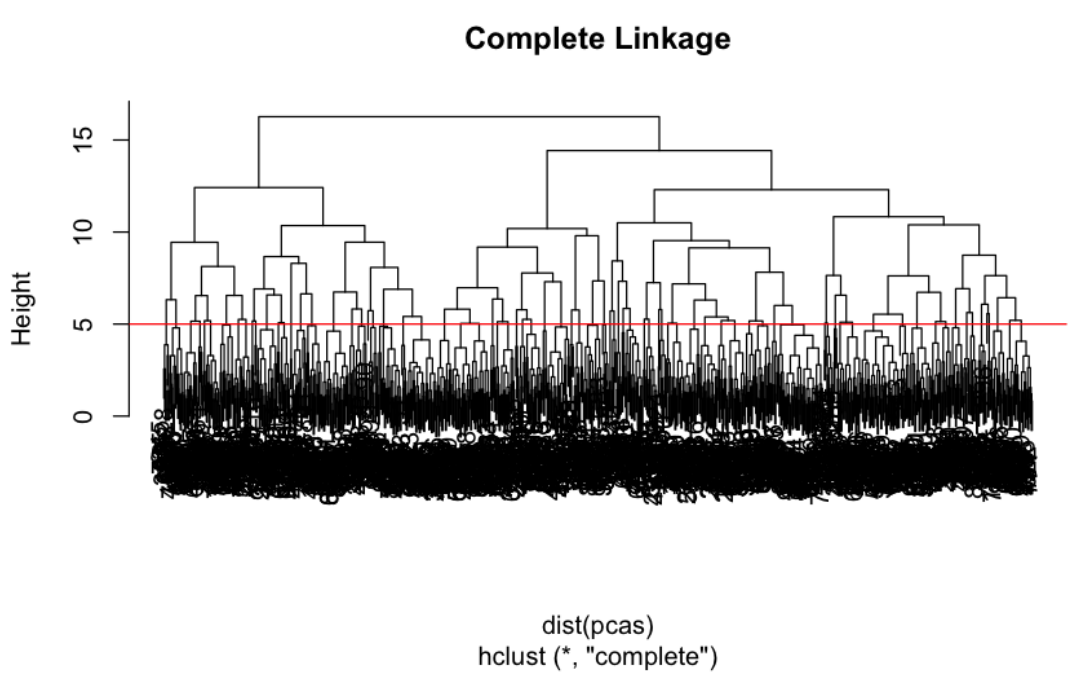

Figure 23: Hierarchical Clustering with Complete Linkage

Next, since the success of hierarchical clustering is highly dependent on which linkage method we use, (Hastie et al., referenced in Lovik 2025), I also fit hierarchical clustering with mean inter-cluster dissimilarity as the score of interest. This measure aims to maximise the average dissimilarity between all members of a group, instead of partitioning based on the absolute maximum value it can produce. The dendogram it produces indicates that the most suitable number of clusters is 18. This is also not a very interpretable number of clusters.

Since k-means clustering suggested a group size of 5 and this is a much more manageable number of groups, I proceed with reporting the results for this number of groups.

Question 7: How large is each group?

The size of groups produced by 5-means clustering is reported below. It seems to split the datapoints into roughly well-balanced groups, as would be expected from this method.

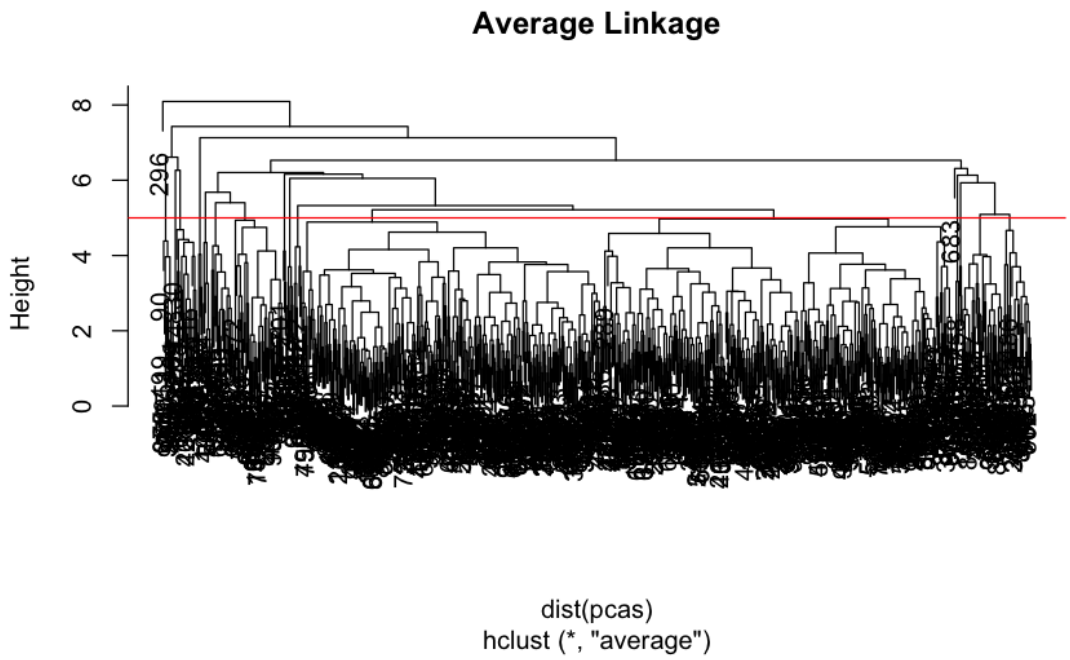

Figure 24: Hierarchical Clustering with Average Linkage

1	2	3	4	5
204	167	198	204	227

Figure 25: K-means clustering group size

Question 8: If you identified more than one technique, what are the main differences between your chosen methods?

The differences between hierarchical and k-means clustering arise from how each algorithm determines how data should be clustered. Their main difference arise from whether the group number must be specified before running the algorithm.

Firstly, hierarchical clustering is group-agnostic, and its tree-like structure arises from the fact that it tries to come up with the most fitting number of groups to cluster the data into. In the agglomerative approach, the algorithm treats every datapoint as its own group at the outset. After this, it merges the most similar datapoints together based on the dissimilarity score that its linkage function specifies. Based on how similar these fused clusters are, it determines where this fusion should be placed on the dendogram (Lovik, 2025) – higher dissimilarity values go higher, lower dissimilarity values go lower. By doing this for every datapoint, we end up with fewer and fewer groups, and the dissimilarity score for each fusion becomes greater and greater. The point at which the greatest "jump" occurs is the ideal number of groups identified on the dendogram by the red line.

On the other hand, k-means requires a pre-determined number of groups to run the algorithm. The scree plot above which helped us determine the best number of k is not a part of the algorithm itself, each individual k-means clustering attempt was fit separately from each other with a different number of pre-defined groups. This is because the k-means algorithm starts by (1) randomly determining 5 vectors which it will treat as the "centroids" for the groups, (2) the datapoints will then be assigned to the group whose centroid has the smallest euclidean distance to them. (3) Once all points have been assigned to a group, the *actual* centroid of the group is determined, and (4) points are re-assigned to their new closest centroid. Steps 3 and 4 are repeated until the group assignments no longer change. (DATAtab, 2021) This results in the 5 groups with the greatest similarity within-groups and dissimilarity between-groups. We can see from the definition of the algorithm, that the number of groups is not informed by the data, rather the data are partitioned so as to fit into a desired number of groups. Also, it is evident from the description of the algorithm that it is sensitive to outliers, as points with values very "far" in the dimensional space than other datapoints can influence where the centroid of the group they are assigned to is placed. This is not the case for hierarchical clustering, as outliers or extreme values do not significantly shift the way groups are assigned to the rest of the data.

In terms of application to this problem, the two methods produced very different groups – as did the different linkage functions for hierarchical clustering.

Question 9: What are the main differences between the groups in terms of personality (give a substantive interpretation of the obtained clustering)?

The boxplots below show how each principal component was assigned to each group. People high on values of the first principal component (the high-achieving personality profile) were most often assigned to group 4, and least often assigned to group 2. The means for this difference in group assignment are significant, whereas for the other three groups this difference does not seem to be. As such, group 1 subjects tend to be much higher in competence, self-discipline and dutifulness and much lower in vulnerability, depression and self-consciousness than people in cluster 4.

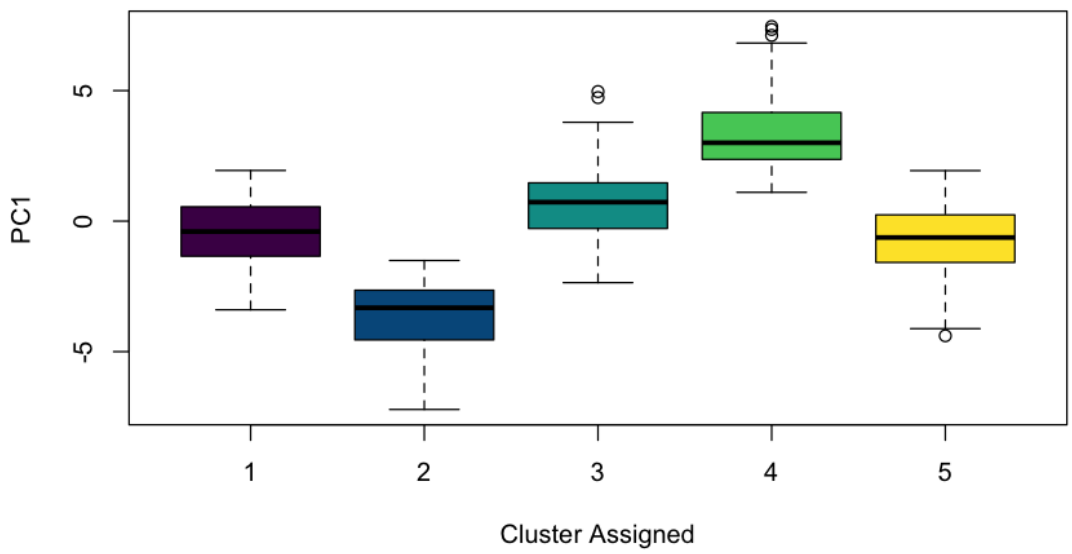

Figure 26: Principal Component 1 group assignments

People scoring high on Principal Component 2 (the buttoned-up characteristics) are most frequently assigned to group 3, and leas likely to be assigned to group 5. This means that personalities in cluster 3 are more likely straightforward, compliant, modest, and have low feelings, excitement-seeking, fantasy and activity than those in group 5.

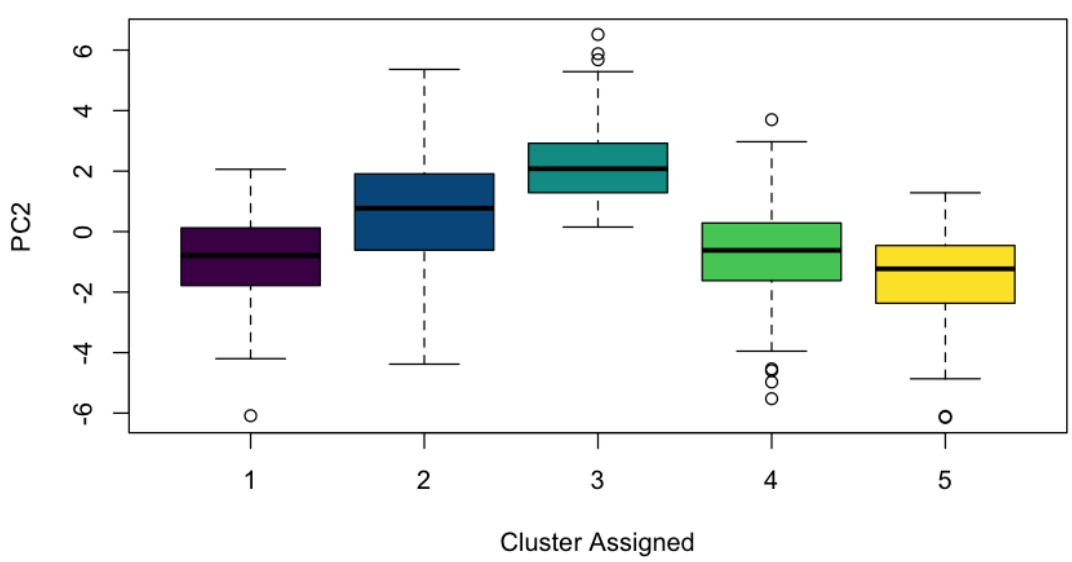

Figure 27: Principal Component 2 group assignments

For people with high scores for Principal Component 3 (the soft-heartedness trait), group 5 seems to be the most frequent cluster, and group 1 the least frequent. Group 5 is also the least frequent group of assignment for those possessing high scores on PC1, and this is logical as some of the most negative loadings on PC3, like self-discipline and achievement striving are some of the most positive loadings on PC1. This seems to indicate that group 5 and group 1 are quite different personality-types from each other, mostly diverging in the extent to which they are geared towards achievement and self-discipline.

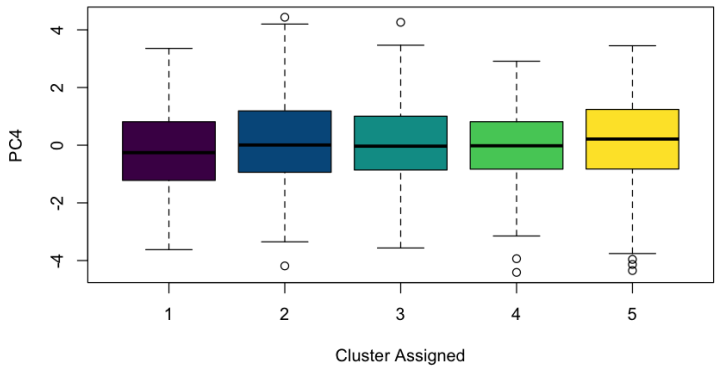

Figure 29: Principal Component 4 group assignments

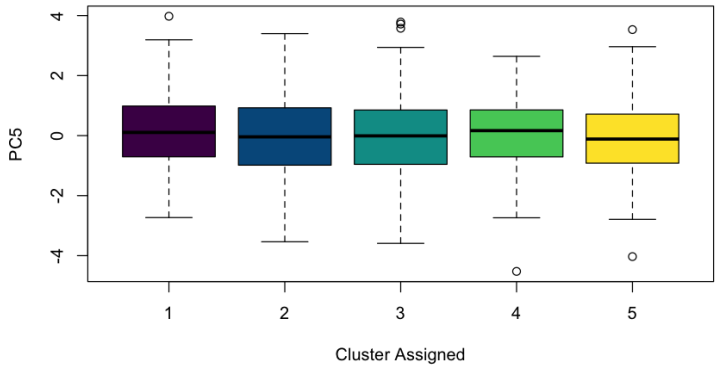

Figure 30: Principal Component 5 group assignments

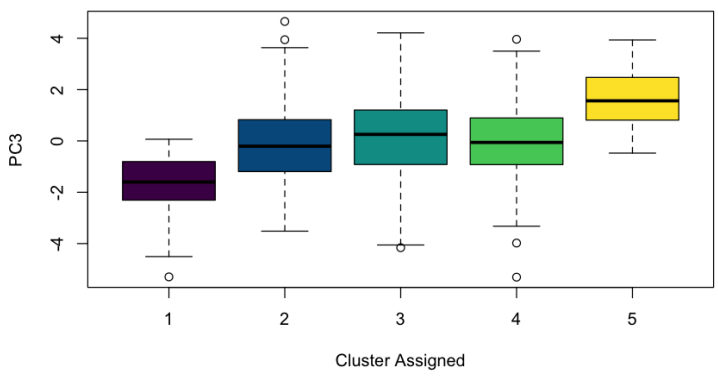

Figure 28: Principal Component 3 group assignments

Principal Components 4 and 5 unfortunately do not differ significantly in their mean group assignment, showing that perhaps five clusters is too many for five principal components. Since the elbow on the scree plot used to select the number of clusters also indicated that k = 2 might be a suitable number of clusters, I re-run the whole analysis with k = 2. Which unfortunately results in even more flat boxplots.

This shows that perhaps a more suitable number of principal components could have been chosen, in order to result in higher between-group heterogeneity. Five principal components were recommended via Horn's Parallel analsis, Velicer's MAP and the "lower" elbow in the Scree plots. This was done in order to maximise the cumulative variance captured by the reduced-dimensionality factors. However, in future analysis, perhaps a smaller number of principal components, for example 3 as recommended by VSS or 4, as indicated by the "steeper" elbow on the Scree plot could yield clusters that are more heterogeneous to each other.

References

Cattell, R. B. (1966). The Scree Test For The Number Of Factors. Multivariate Behavioral Research, 1(2), 245–276. https://doi.org/10.1207/s15327906mbr0102_10

DATAtab, (2021, January 13). *k-means Cluster Analysis* [Video]. Youtube. https://www.youtube.com/watch?v=GZj6ikx8PAc

Friedman, J. H. (2001). Greedy Function Approximation: A Gradient Boosting Machine. The Annals of Statistics, 29(5), 1189–1232. http://www.jstor.org/stable/2699986

Funder, DC, Ozer, DJ. (2019). Evaluating Effect Size in Psychological Research: Sense and Nonsense. Advances in Methods and Practices in Psychological Science. 2019;2(2):156-168. https://doi.org/10.1177/2515245919847202

Horn, J.L. A rationale and test for the number of factors in factor analysis. Psychometrika 30, 179–185 (1965). https://doi.org/10.1007/BF02289447

Ledesma, R. D., Valero-Mora, P. M., & Macbeth, G. (2015). The Scree Test and the Number of Factors: a Dynamic Graphics Approach. The Spanish Journal of Psychology, 18. https://doi.org/10.1017/sjp.2015.13.

Revelle, W., Rocklin, T. 1979, Very Simple Structure: an Alternative Procedure for Estimating the Optimal Number of Interpretable Factors, Multivariate Behavioral Research, 14, 403-414. https://doi.org/10.1207/s15327906mbr1404_2

VSS: Apply the very simple structure, map, and other criteria to determine the appropriate number of factors. RDocumentation. (n.d.). https://www.rdocumentation.org/packages/psych/versions/2.5.3/topics/VSS

Lovik, A. (2025). Week 10 - Unsupervised Learning 2 (Clustering) [Presentation Slides]. Leiden University. https://brightspace.universiteitleiden.nl/d2l/le/lessons/310446/topics/3229262

XGBoost parameters. XGBoost 3.1.0 Dev Documentation. (n.d.). https://xgboost.readthedocs.io/en/latest/parameter.html